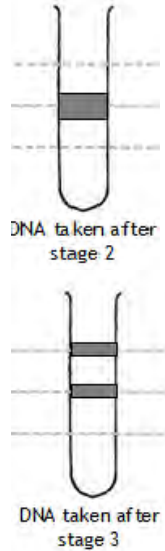
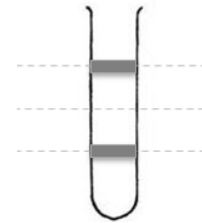


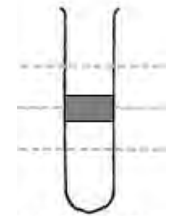
Question number	Answer	Additional guidance	Mark
8(a)	An explanation that includes the following points: • form base pairs to hold the two {polynucleotide / strands} together (1) • because of the number of hydrogen bonds that each type of base can form (1) • a {large base / purine} bonds with a {small base / pyrimidine} (1) • so that the strands are parallel (1)	ACCEPT adenine and thymine form two hydrogen bonds <u>and</u> guanine and cytosine form three hydrogen bonds ACCEPT single ring bonds to double ring	(3)

Question number	Answer	Additional guidance	Mark
8(b)(i)	• 6.6×10^2 (1)	ACCEPT 6.596×10^2	(1)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	- mass of DNA after three divisions calculated (1) $\{4 / 4.1 / 4.14 / 4.142\} \times 10^{-14}$ (1)	$(3.1 \times 10^9 \times 8) \quad 24.8 \times 10^9$ NB Correct answer but wrong standard form = 1 mark ECF $\{2 / 1.6 / 1.55\} \times 10^{-14} = 1$ mark	(2)

Question number	Answer	Additional guidance	Mark
8(c)(i)	A diagram that shows the following points: - a band the same width as stage 1 in the middle of the tube (1) - bands drawn at the top and middle of tube (1) - both bands half the width of stage 1 (1)	 NB For mp 3, if 1 band is drawn allow 1 mark if same width as the band in stage 1 level and above it	(3)

Question number	Answer	Additional guidance	Mark
8(c)(ii)	A diagram that shows the following points: - bands drawn at the top and bottom of tube (1) - both bands half the width of stage 1 (1)	 DNA taken after stage 2 NB If 1 band is drawn allow 1 mark if same width as the band in stage 1 and above it	(2)

Question number	Answer	Additional guidance	Mark
8(c)(iii)	A diagram that shows the following points: - one band drawn in the middle (1) - of similar thickness to band in stage 1 (1)	 DNA taken after stage 2	(2)